# Choosing a statistical test

Work through the four questions, then read across the row. The test is decided by the design and the type of outcome — never by the result you would like to get.

### FIRST: WHAT KIND OF VARIABLE IS IT?

| Nominal | Ordinal | Interval | Ratio |
|---|---|---|---|
| Names of categories, no order at all. You can only count how many fall in each. *sex, tooth type, filling material A / B / C, alive or dead* | Categories with a clear order, but the steps between them are not equally sized. *pain 0–3, plaque index, tumour stage, "agree / neutral / disagree"* | Numbers with equal steps, but zero is arbitrary — differences make sense, ratios do not. *temperature in °C, calendar year, pH* | Equal steps and a real zero, so ratios are meaningful: 4 mm is twice 2 mm. *probing depth, age, number of teeth, cost in CHF* |

On this sheet, **interval and ratio together are called "continuous"**. Ordinal data are usually handled with the same tests as skewed continuous data. Nominal data with two levels (yes / no) have their own tests. Choosing a test for a higher scale than your data actually have is a mistake — averaging a pain score of 0–3 assumes steps you do not have.

| Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|
| **How many predictors?** One → Part A · Several, adjusted for each other → Part B | **Difference or association?** Comparing groups · or measuring how two things move together | **Independent or paired?** Different subjects in each group · or the same subject measured twice | **What kind of outcome?** Continuous · ordinal · yes/no · count · time to event |

## Part A · One predictor *two or more groups, nothing adjusted for*

### DIFFERENCE — INDEPENDENT GROUPS

| OUTCOME | GROUPS | TEST |
|---|---|---|
| Continuous, roughly normal | 2 | `t-test` (Welch's version by default) |
| Continuous, roughly normal | >2 | `One-way ANOVA` |
| Continuous but skewed, or ordinal | 2 | `Mann–Whitney U test` |
| Continuous but skewed, or ordinal | >2 | `Kruskal–Wallis test` |
| Yes / no, or unordered categories | any expected count <5 | `Fisher's exact test` |
| Yes / no, or unordered categories | all expected counts ≥5 | `Chi-square test` |
| Time to an event with drop-outs / different follow-up | 2 or more | `Kaplan–Meier curve` to describe, then `log-rank test` to compare |

### DIFFERENCE — PAIRED OR REPEATED MEASUREMENTS

| OUTCOME | MEASUREMENTS | TEST |
|---|---|---|
| Continuous, roughly normal | 2 | `Paired t-test` |
| Continuous, roughly normal | >2 | `Repeated-measures ANOVA` |
| Continuous but skewed, or ordinal | 2 | `Wilcoxon signed-rank test` |
| Continuous but skewed, or ordinal | >2 | `Friedman test` |
| Yes / no | 2 | `McNemar's test` |
| Yes / no | >2 | `Cochran's Q test` |

### ASSOCIATION BETWEEN TWO VARIABLES

| VARIABLES | QUESTION | METHOD |
|---|---|---|
| Both continuous, roughly normal | do they move together? | `Pearson's correlation` (r) |
| Skewed or ordinal | do they move together? | `Spearman's correlation` ($r_s$) |
| Two raters, categorical ratings | do they agree? | `Cohen's kappa` |

## Part B · Several predictors *regression — the effect of one variable, adjusted for the others*

**ONE ROW PER SUBJECT — INDEPENDENT OBSERVATIONS**

| OUTCOME | EXTRA CONDITION | MODEL |
|---|---|---|
| Continuous | residuals roughly normal | `Linear regression` |
| Continuous | residuals skewed | `Linear regression` on transformed outcome, or with robust SEs |
| Ordered categories | — | `Ordinal logistic regression` |
| Yes / no | — | `Binary logistic regression` |
| Unordered categories >2 | — | `Multinomial logistic regression` |
| Count or rate | spread ≈ mean | `Poisson regression` |
| Count or rate | spread > mean (the usual case) | `Negative binomial regression` |
| Time to an event | — | `Cox proportional-hazards regression` |

**SEVERAL ROWS PER SUBJECT — REPEATED OR CLUSTERED DATA**

| OUTCOME | EXTRA CONDITION | MODEL |
|---|---|---|
| Continuous | residuals roughly normal | `Linear mixed-effects model` or GEE |
| Continuous | residuals skewed | `Linear mixed-effects model` on transformed outcome |
| Ordered categories | — | `Ordinal mixed model` or GEE |
| Yes / no | — | `Logistic mixed model` or GEE |
| Count or rate | — | `Poisson / negative binomial mixed model` or GEE |
| Time to an event | — | `Cox regression` with cluster-robust standard errors |

### SIX RULES OF THUMB

1. **Design first.** If the same subject appears twice in your data, you are in a paired or clustered row. Ignoring that is the most common serious mistake.
2. **Normality applies to the right thing.** For a t-test it is the outcome within each group; for a regression it is the residuals. With large samples both are forgiving.
3. **Fisher vs. chi-square** is decided by the *expected* cell counts, not by the total sample size.
4. **Counts are not continuous.** Poisson assumes the spread equals the mean. Real data are almost always more spread out — then use negative binomial. If follow-up time differs between subjects, add an offset and model a rate.
5. **Part A assumes no confounding.** A single-predictor test takes the groups as they come. If a third variable differs between the groups and also affects the outcome, move to an adjusted analysis in Part B. Budget roughly **10 patients per variable** you put in the model.
6. **Check interactions.** A model with main effects only assumes the effect is the same in every subgroup. Test the few interaction terms that are clinically plausible.